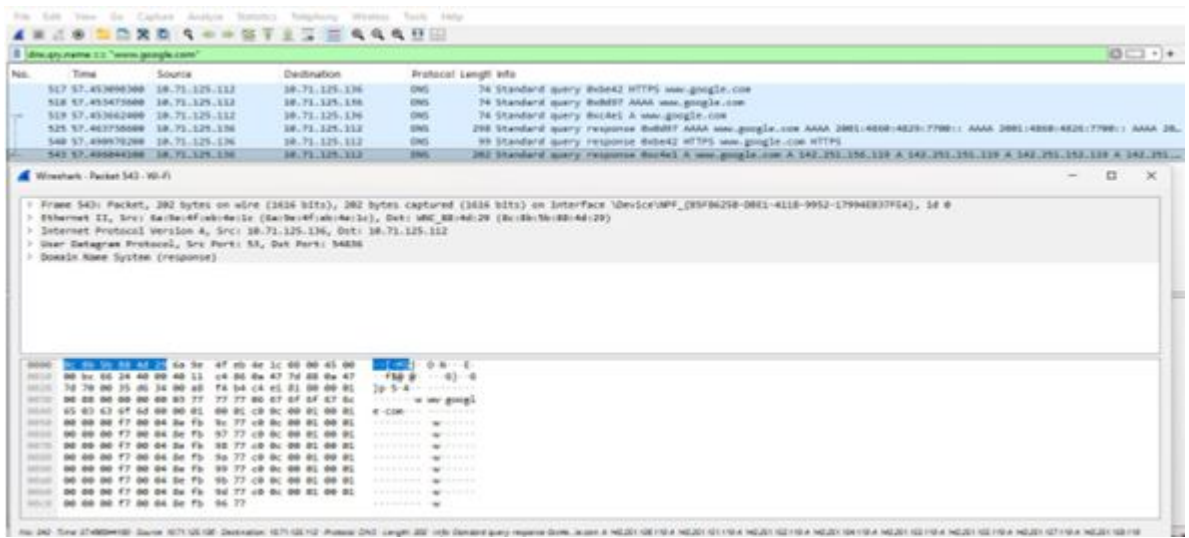


CO5-AT1 CONSTRAINT BASED
PROBLEM SOLVING

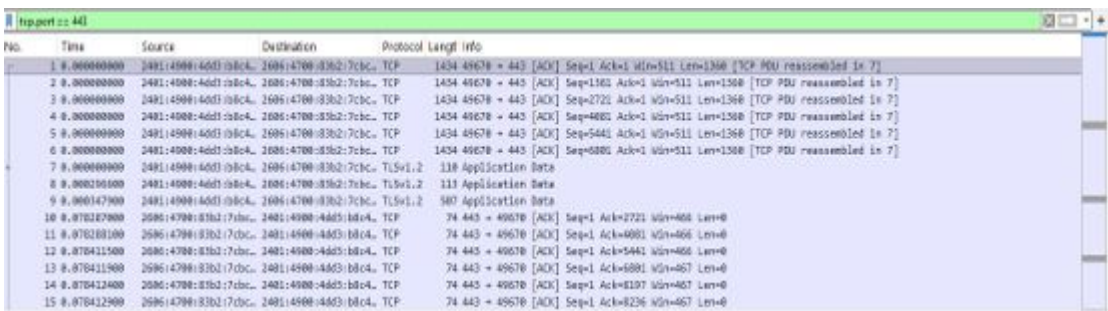
NAME:madhumitta Ch S
REG NO:1921260101

Annex: 3. Constraint-Based Problem Solving 1. DNS Resolution – Google



Packet Number: 543 Wireshark Filter: `dns.qry.name == "www.google.com"`

2. HTTPS Web Access – Wikipedia



Protocol: TCP
filter used: `tcp.port == 443`

3. FTP Traffic Challenge – Real FTP Test Service FTP

uses two connections:

- Port 21 → FTP control connection 
- Port 20 → FTP data connection (in traditional active FTP)

The control connection carries commands such as login, directory listing requests, and file-transfer commands.

Wireshark filter:

```
tcp.port == 21
```

Option: Port 21

4. SMTP Traffic Identification – Mailtrap Sandbox

SMTP (Simple Mail Transfer Protocol) is used to send/transmit outgoing emails.

The other options are:

- SMTP → Sends outgoing email
- POP → Retrieves/downloads email
- DNS → Resolves domain names
- SNMP → Network management